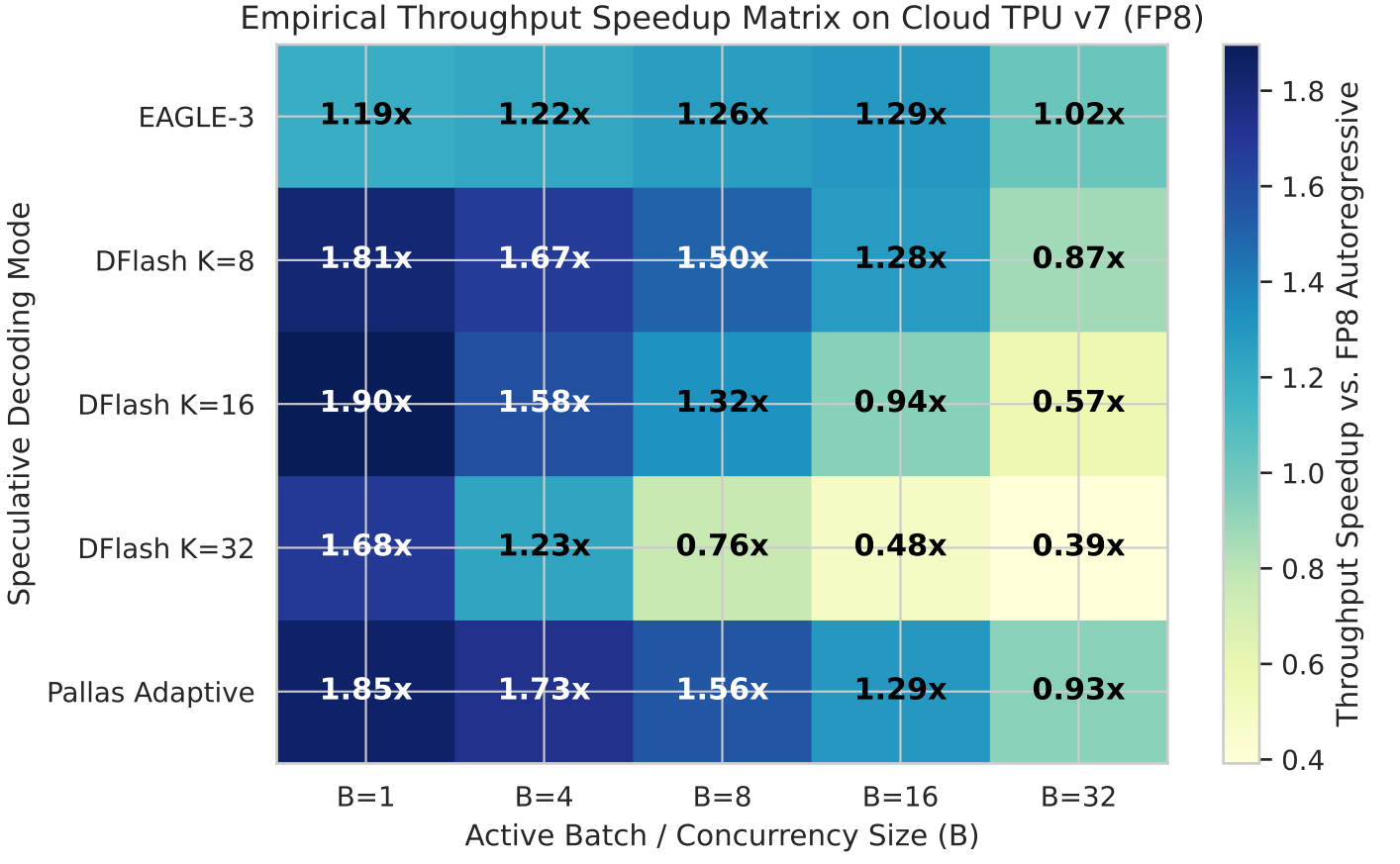
Empirical Throughput Speedup Matrix on Cloud TPU v7 (FP8)
Speculative Decoding Mode
EAGLE-3
DFlash K=8
DFlash K=16
DFlash K=32
Pallas Adaptive
1.19x
1.22x
1.26x
1.29x
1.02x
1.81x
1.67x
1.50x
1.28x
0.87x
1.90x
1.58x
1.32x
0.94x
0.57x
1.68x
1.23x
0.76x
0.48x
0.39x
1.85x
1.73x
1.56x
1.29x
0.93x
B=1
B=4
B=8
B=16
B=32
Active Batch / Concurrency Size (B)
Throughput Speedup vs. FP8 Autoregressive
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4